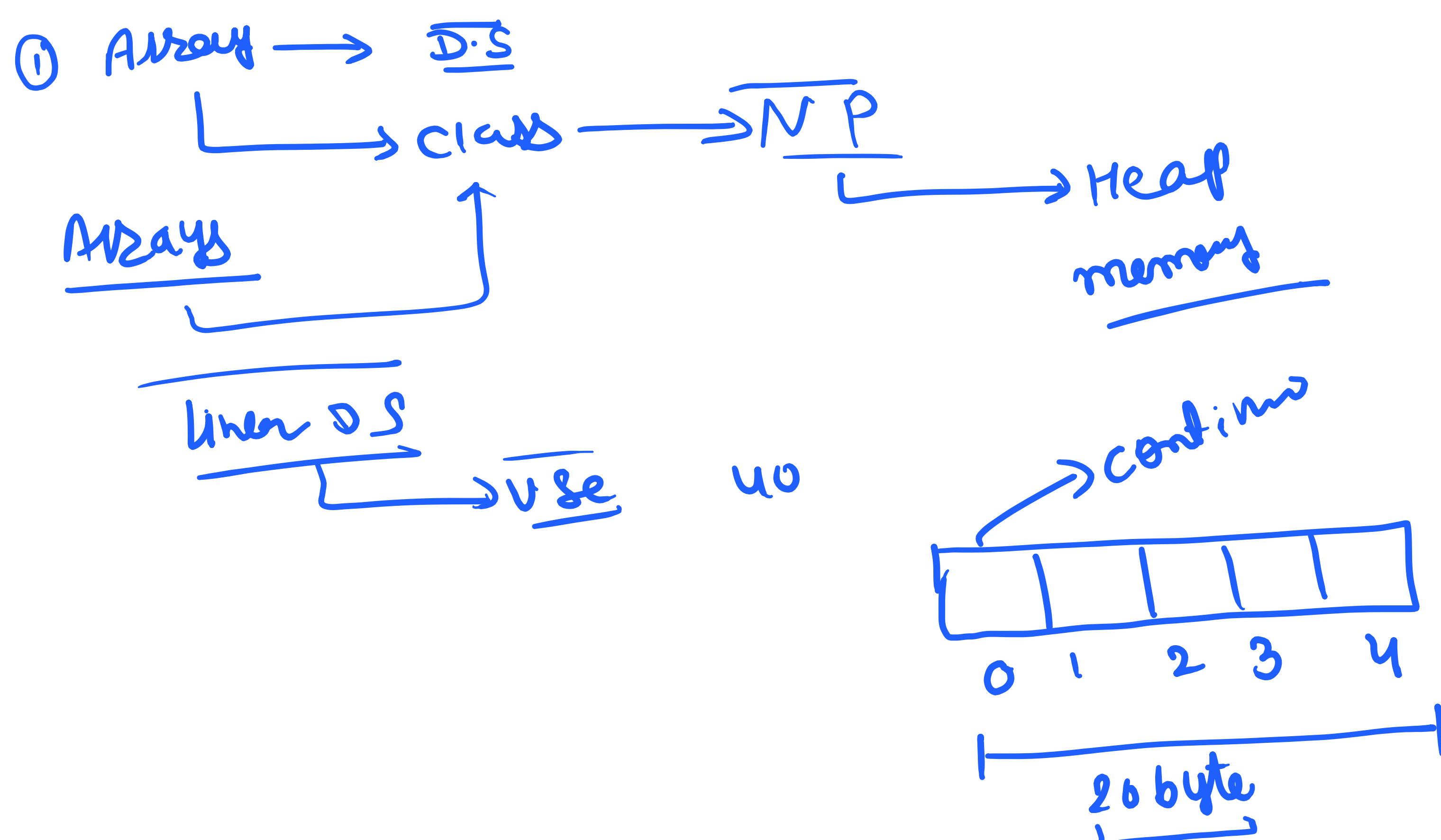
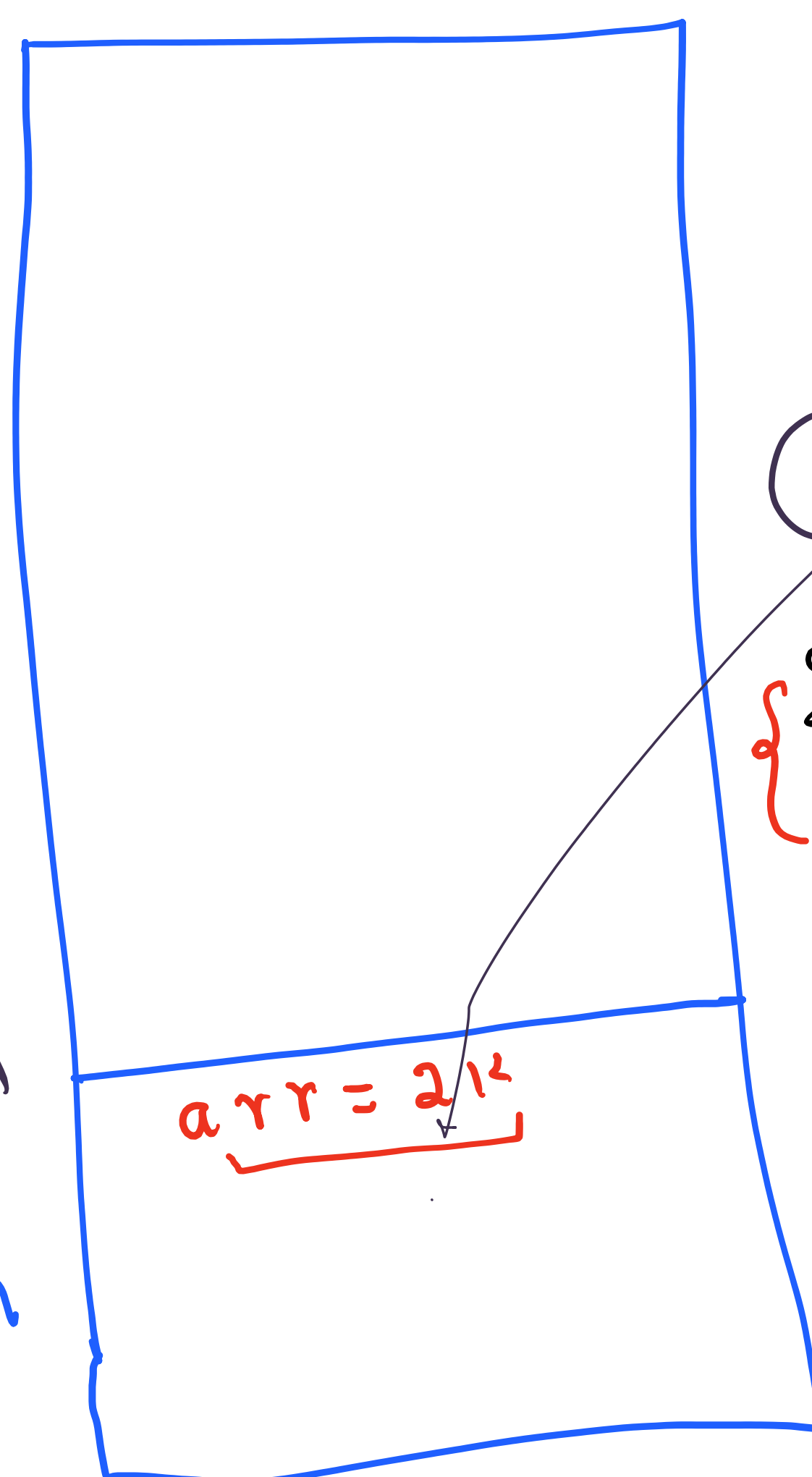




```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    int[] arr = new int[5];  
    System.out.println(arr);  
}
```

Sys0(arr0)
Sys0(arr1)
Sys0(arr2)
Sys0(arr3)
Sys0(arr4)

arr0=5
arr1=7
arr2=9
arr3=7
arr4=1
main



Why zero boxing

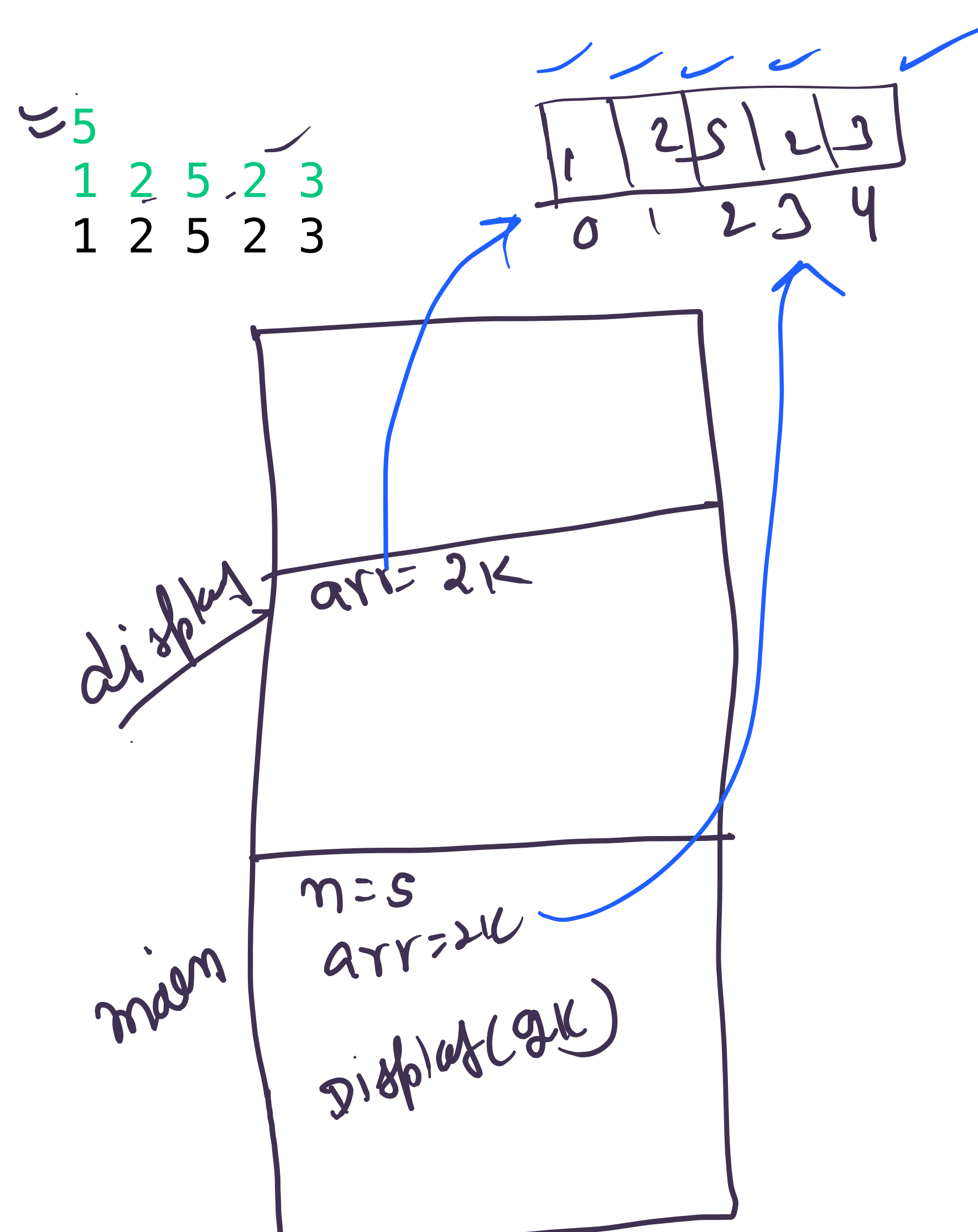
0 1 2 3 4

0 0 5 0 0

arr0=5

2K + 2KSize
2K + 2x4 = 2008
arr0 = 5
arr1 = 7

```
import java.util.Scanner;  
public class User_input {  
    public static void main(String[] args) {  
        // TODO Auto-generated method stub  
        Scanner sc = new Scanner(System.in);  
        int n = sc.nextInt();  
        int[] arr = new int[n];  
        for (int i = 0; i < arr.length; i++) {  
            arr[i] = sc.nextInt();  
        }  
        display(arr);  
    }  
    public static void display(int[] arr) {  
        for (int i = 0; i < arr.length; i++) {  
            System.out.print(arr[i] + " ");  
        }  
    }  
}
```

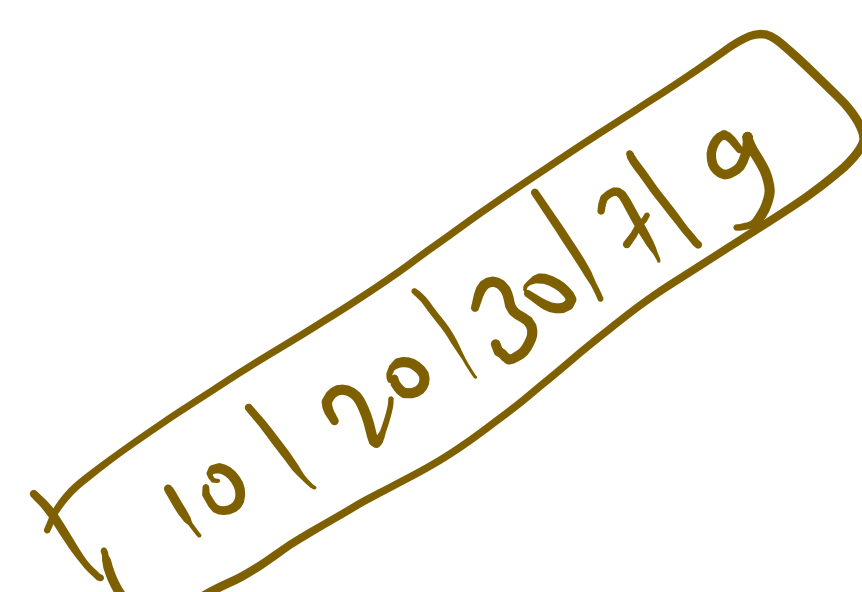


```
int[] arr = { 10, 20, 30, 7, 9 };
```

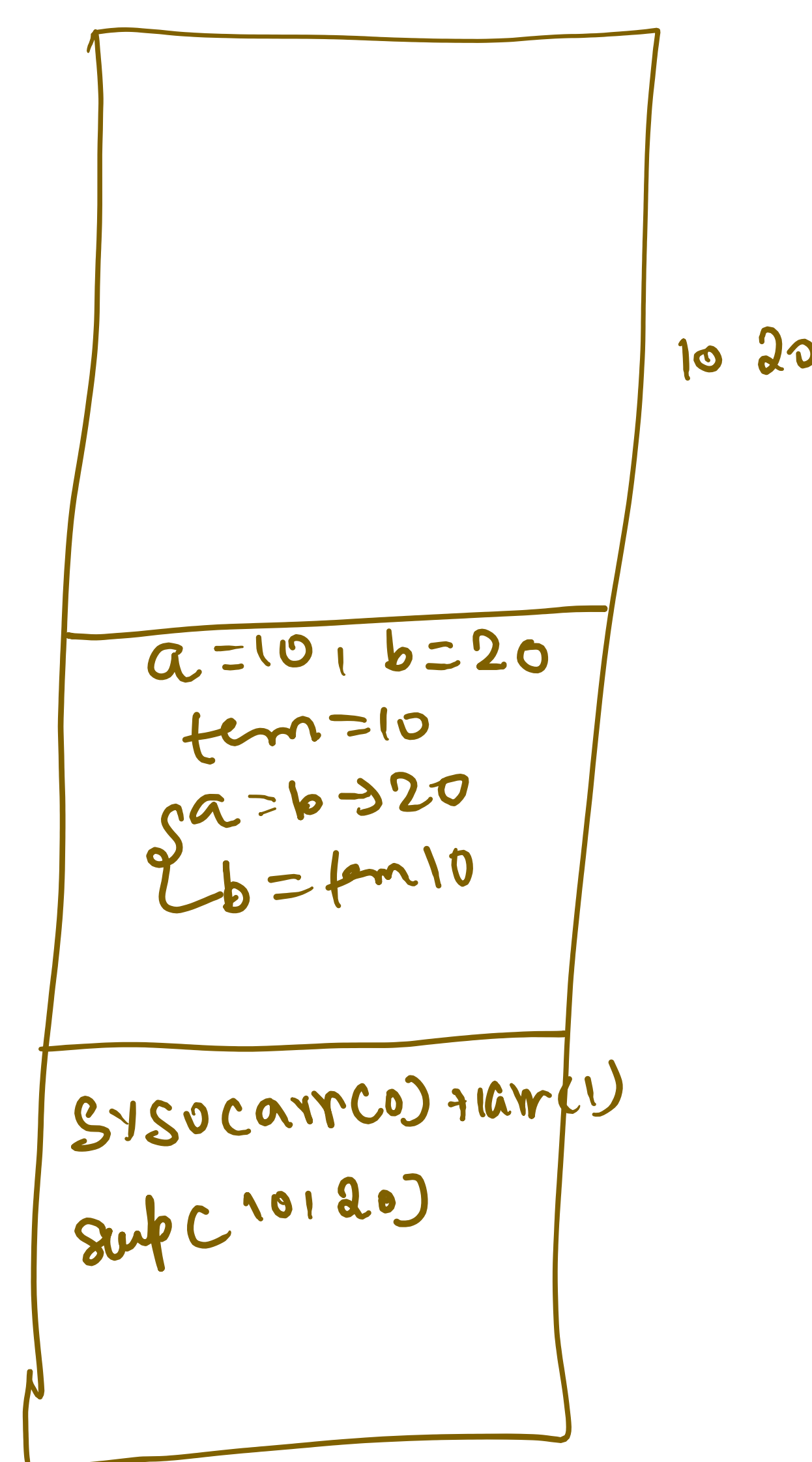
a = 10
b = 30

temp = 9
a = b
b = temp

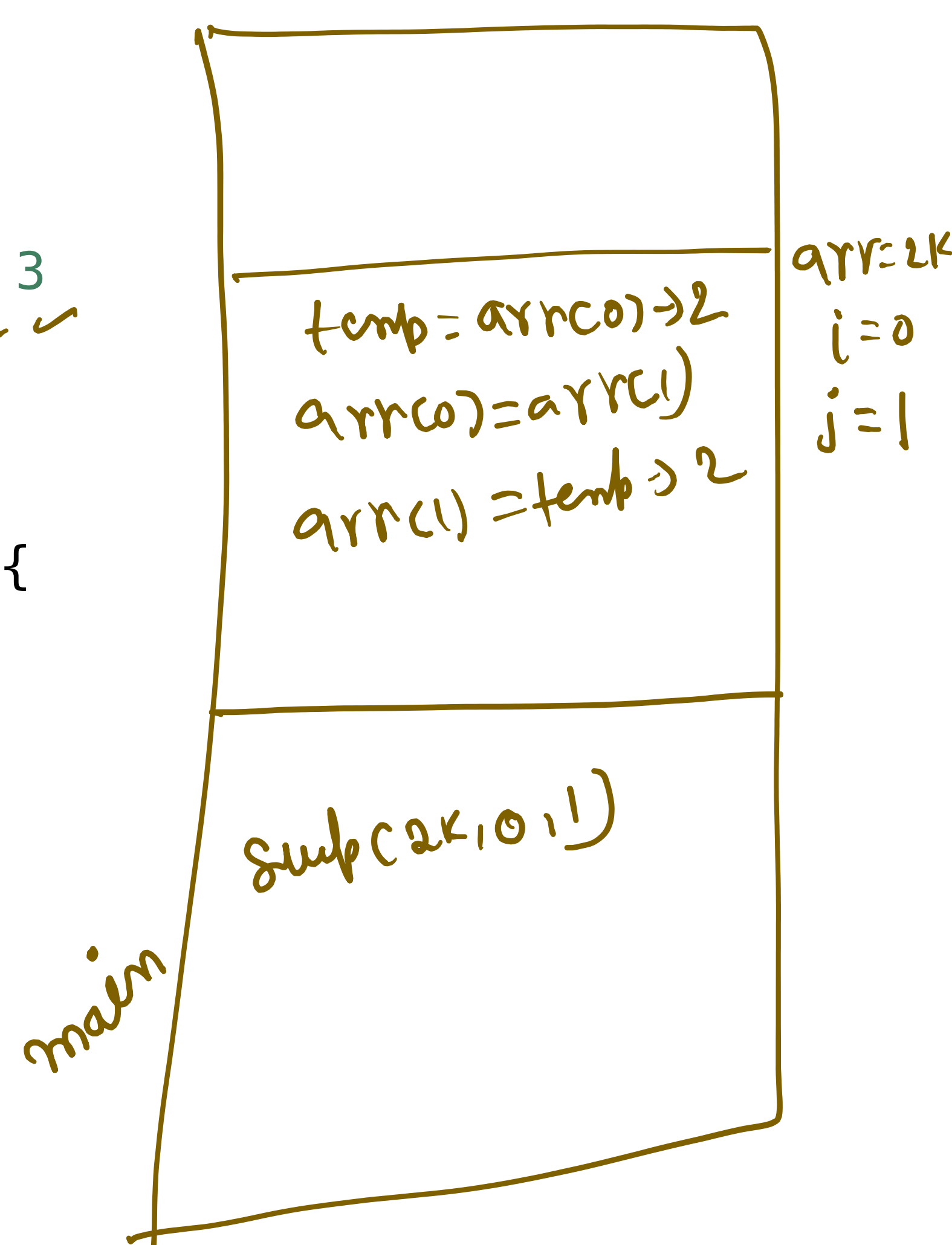
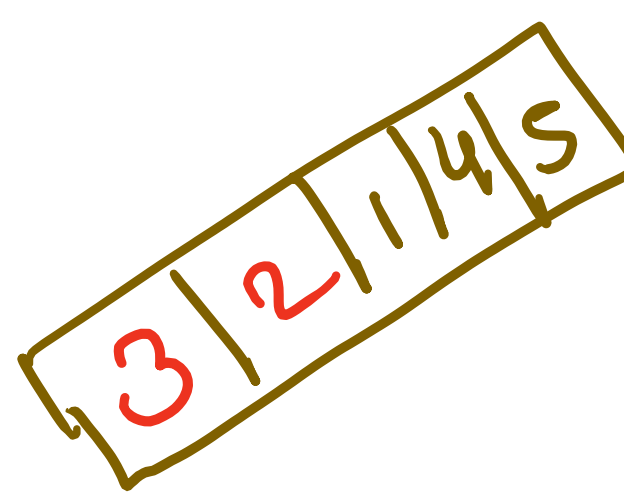
```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    int[] arr = { 10, 20, 30, 7, 9 };  
    // int[] arr1 = new int [] { 10, 20, 30, 7, 9 };  
    System.out.println(arr[0] + " " + arr[1]);  
    swap(arr[0], arr[1]);  
    System.out.println(arr[0] + " " + arr[1]);  
}  
public static void swap(int a, int b) {  
    int temp = a;  
    a = b;  
    b = temp;  
}
```



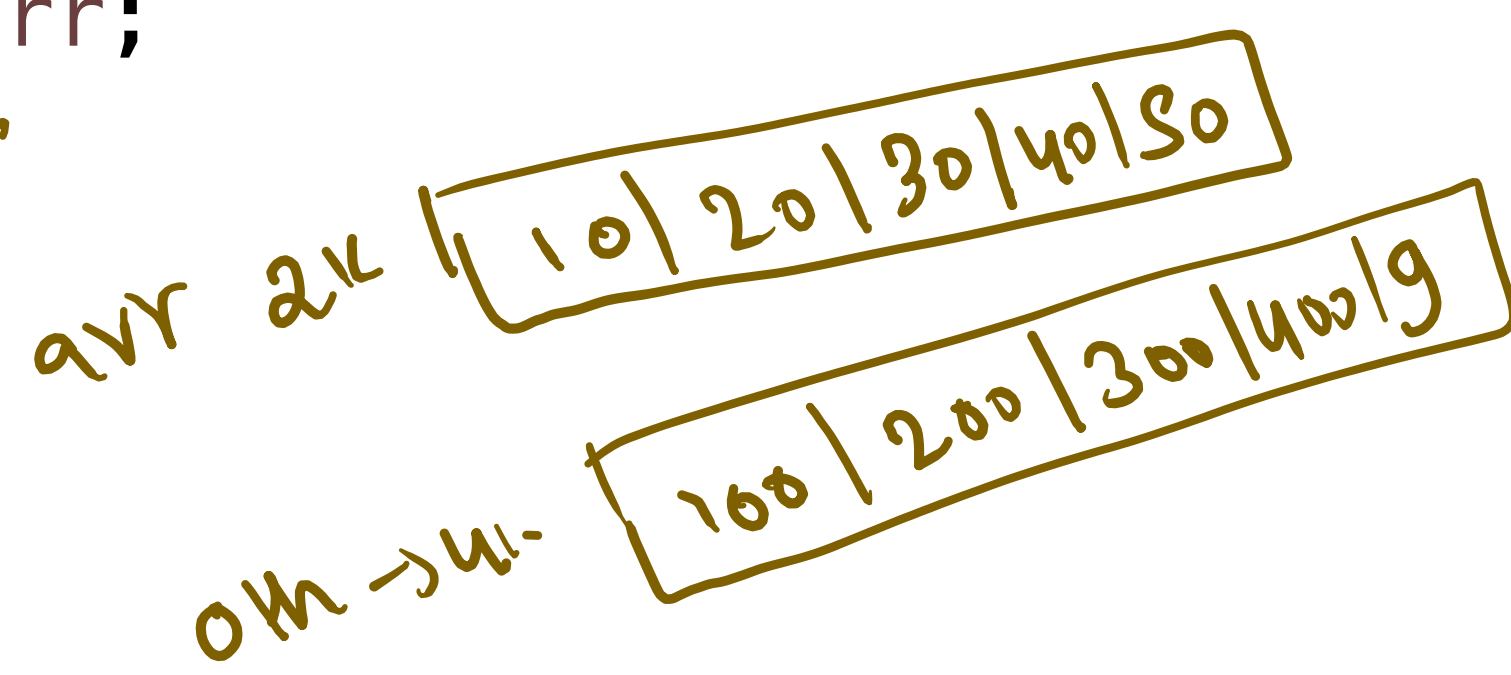
main



```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    int[] arr = { 2, 3, 1, 4, 5 };  
    System.out.println(arr[0] + " " + arr[1]);  
    swap(arr, 0, 1);  
    System.out.println(arr[0] + " " + arr[1]);  
}  
public static void swap(int[] arr, int i, int j) {  
    int temp = arr[i];  
    arr[i] = arr[j];  
    arr[j] = temp;  
}
```



```
public static void main(String[] args) {  
    // TODO Auto-generated method stub  
    int[] arr = { 10, 20, 30, 40, 50 };  
    int[] other = { 100, 200, 300, 400, 9 };  
    System.out.println(arr[0] + " " + other[0]);  
    swap(arr, other);  
    System.out.println(arr[0] + " " + other[0]);  
}  
public static void Swap(int[] arr, int[] other) {  
    // TODO Auto-generated method stub  
    int[] temp = arr;  
    arr = other;  
    other = temp;  
}
```



main

